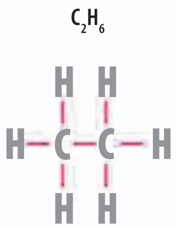
Alkanes, alkenes, and alkynes

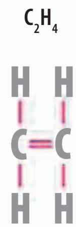
Chained hydrocarbons form families according to how their carbon atoms are bonded. Hydrocarbons with single bonds are called alkanes. Chains with at least one double bond are alkenes. A triple-bonded compound is an alkyne.

Suffix	Contains
ane	carbon–carbon single bonds
ene	carbon–carbon double bonds
yne	carbon–carbon triple bonds

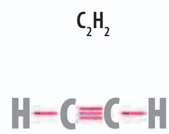
◁ **Suffix**
Compounds are given a suffix to show which family they belong to.



◁ **Ethane**
The single bond in ethane makes this hydrocarbon relatively stable and unreactive.



◁ **Ethene**
The double bond makes ethene more flammable than ethane.

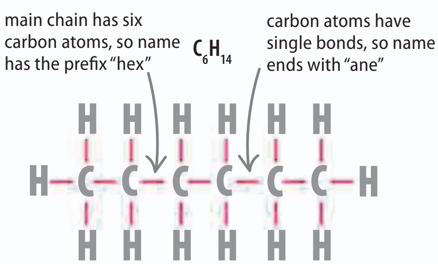


◁ **Ethyne**
The triple bond is very unstable. Ethyne and all alkynes are highly flammable and reactive.

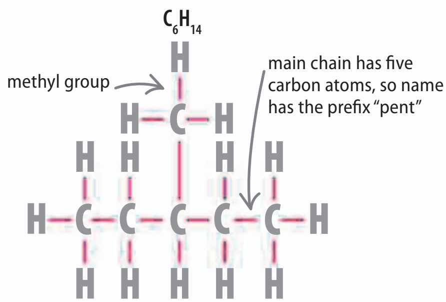
reactivity increasing

Isomers

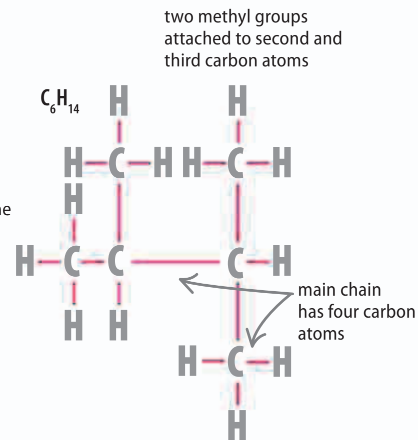
Aliphatic compounds can have the same formula—the number of carbon and hydrogen atoms—but be arranged in different ways. These similar compounds are known as isomers. Side branches change the way isomers behave, making them react differently and have different melting and boiling points.



△ **Hexane**
This liquid alkane has six carbon atoms in a single chain. It has a total of four isomers and its main use is in petrol.



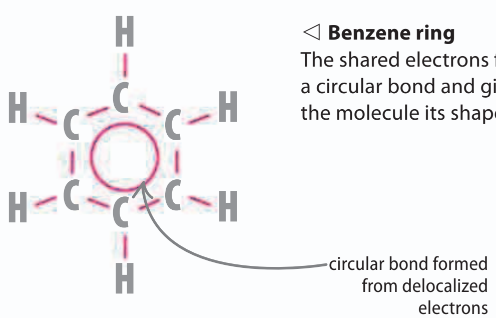
△ **Methylpentane**
The longest chain in this compound is a pentane. A methyl group (side chain with one carbon) adds the sixth carbon atom.



△ **2,3-Dimethylbutane**
The longest chain in this isomer is a butane. Two methyl groups are attached to second and third carbon atoms in the butane.

Aromatics

Hydrocarbons can also form ringed molecules called aromatics. The simplest of these is benzene (C_6H_6), which has six carbon atoms linked with alternating single and double bonds. The electron pairs forming the three double bonds are free and shared between all six carbon atoms, forming a ring-shaped "delocalized" bond.



◁ **Benzene ring**
The shared electrons form a circular bond and give the molecule its shape.